Assessing the influence of variable annual sampling effort on official national forest inventory statistics

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Abstract

In this study we…

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1. Introduction

The USFS Forest Inventory and Analysis (FIA) Program surveys and analyzes the forests of the United States through a nationwide network of permanent sample locations. These plot locations are distributed across all lands with a spatial sampling intensity of about 1 location per 2400 ha according to an unaligned, quasi-systematic equal probability sample design. This design is spatially dispersed with deviation from a regular grid to avoid alignment of sample locations with periodic population features. Statistical inference on forest population status and change parameters is enabled by measurements of forest vegetation traits made at these locations by field crews.

Historically, FIA sample locations in each state were visited periodically as time, budgets, and workforce capacity allowed. This approach offered logistical flexibility but complicated analysis and reporting timelines. In the early 2000's the program shifted from periodic to annual inventory protocols calling for approximately equal numbers of plot measurements each year on schedules intended to be fixed and consistent within states. Under this annual inventory, each plot location in a state is assigned to one of a set of panels, and all plot locations assigned to a panel are designated for measurement in the same year.

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The panels in a state together comprise an inventory cycle, and inventory cycles range in length from 5 or 7 years in the eastern US to 10 years in the west. Once the final panel in an inventory cycle is complete, remeasurements at sample locations in the first panel begin the following year. For more information regarding the FIA sample design see [1, 2, Chapter 2].

In general, FIA requires that a panel be complete before the associated tabular data are pushed to its public-facing databases, which the program endeavors to do within 1 year of panel completion. Similarly, FIA requires a complete inventory cycle within a state to publish official state-scale estimates of forest parameters in fulfillment of Congressional reporting obligations. In practice, however, adherence to the annual design described above has proven very challenging or impossible for the FIA Program. Instead, delayed and skipped field measurements cause unintentional lengthening of inventory cycles and/or survey nonresponse [see, e.g., 3], at rates that vary regionally and over time.

The underlying causes of delayed and skipped field measurements are varied, and can be chronic (e.g., denied access to private lands [4], logistical or administrative challenges in remote regions [5], persistent workforce and budgetary constraints [6]) or can result from discrete events (e.g., the COVID19 pandemic [7]). Late measurements of temporally dynamic variables can induce a "time lag" bias in point estimates when the time component of the estimand is not adjusted accordkingly. Further, survey nonresponse can introduce nonresponse bias to estimates of population parameters when response propensity is correlated with the attribute of interest. Better understanding of patterns in late and skipped measurements [6], and the investigation of new sampling and estimation techniques to ameliorate the statistical consequences of nonresponse [8, 9] are active research topics, but beyond the scope of the present study.

Of immediate practical concern for FIA and its external partners are the strain on field staff and delays to tabular data publication that result from delayed field measurements. Methods have been proposed to control the measurement delay process by disaggregating FIA's annual panel structure into subpanels while still preserving the spatial dispersion of the FIA sample network within field season years. Disaggregation of panels into subpanels enables intentional alignment of workloads with anticipated data acquisition workforce capacity in each field season year through the application of variable annual field sampling effort within individual states.

Though the logistical advantages of variable annual effort (VAE) are profound (i.e., it can help ensure the long-term viability of the inventory in the context of funding and staffing volatility), the statistical consequences of the redistribution of partial panels within inventory cycles, and of extended and variable inventory cycle lengths, are unknown across states and epochs. In particular, a need exists to understand and communicate to stakeholders how adoption of VAE protocols may influence the statistical properties of forest parameter estimates published by FIA in fulfillment of Congressional reporting imperatives.

Therefore, our research objectives in this study are to:

1. define three VAE regimens with respect to subpanels (fast and slow start, dropped subpanels, and extended and shortened cycle lengths) intended to be representative of circumstances FIA could conceivably encounter in program administration
2. simulate the influence of those three VAE regimens on state-scale, inventory-cycle wide estimates of total net merchantable timber volume for three US states with distinct forest types, ecological attributes, and FIA inventory cycle lengths.
3. visualize and discuss simulation results in the context of implications for FIA program operations and the management of US forests.

2. Data & Methods

3. Discussion

4. Conclusion

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